Varunprasaath R. Selvaraju, Ph.D.

(+1) 678-207-9914 | varunprasaath@gmail.com | linkedin.com/in/varunprasaath-rs |

PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years in polymer/materials characterization, DOE-guided process development, and structure-property analysis. Currently develops bio-based polymer process windows and uses FT-IR, GPC/SEC, DSC/TGA, NMR, UV-Vis, and structured data to translate formulation/process variables into reproducible specifications.

CORE COMPETENCIES

Polymer Characterization | Process-Structure-Property Analysis | DOE / SPC | GPC/SEC / FT-IR / DSC/TGA | Formulation Process Development | High-Throughput Workflow Ramp | Technical Documentation | Cross-Functional Scientific Communication

WORK EXPERIENCE

University of Colorado Boulder
July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided bio-based polymer process development, linking formulation and process variables to FT-IR, thermal, GPC/SEC-adjacent, and material-quality readouts for scale-up-ready specifications.
- Develop SOP-style records and technical reports that connect polymer characterization data with reproducible process decisions.

University of Colorado Boulder
July 2021 - July 2025

Graduate Researcher

- Developed and characterized functional organic material variants with FT-IR, UV-Vis-NIR, NMR/GPC, TGA/DSC, electrochemistry, and MS to connect structure, purity, stability, and performance.
- Analyzed process-structure-property relationships across 15+ materials and translated complex datasets into design recommendations and peer-reviewed outputs.

Georgia Institute of Technology
Aug 2019 - July 2021

Graduate Researcher

- Co-led AFOSR-funded thin-film coating process development, connecting CVD/surface-preparation conditions to morphology, interface quality, optical performance, and reproducibility.

ETH Zurich
May 2017 - July 2017

Research Assistant

- Developed CVD polymer thin-film deposition workflows using DOE and in-situ QCM-D monitoring to quantify growth kinetics and process controls.

University of Colorado Boulder
2021 - 2025

Laboratory Operations Manager

- Managed characterization/process-development lab operations for a 15-member group; implemented calibration, SOP, and safety systems that improved instrument uptime by 30%.

Georgia Tech
2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Bio-Based Polymer Scale-Up Process Current

Built DOE-guided polymer process windows and material-quality specifications that translate analytical readouts into reproducible scale-up decisions.

DOE, SPC, polymer characterization, FT-IR, thermal analysis

Electrochromic Material Reliability Study Chemistry of Materials 2025

Mapped structure-property-stability relationships across functional organic materials using multi-technique characterization.

FT-IR, UV-Vis-NIR, CV, TGA/DSC, NMR, GPC, MS

AFOSR Defense Thin-Film Coatings DoD Funded

Developed controlled thin-film coating workflows and process-property relationships for high-performance materials.

CVD, QCM-D, surface prep, thin films, DOE

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder 2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc) 2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Polymer & Materials Characterization: GPC/SEC, FT-IR, UV-Vis-NIR, TGA, DSC, NMR, QCM-D, ellipsometry, LC-MS, GC-MS, structure-property-performance analysis
Process & Formulation Development: DOE, SPC, process windows, scale-up specifications, reproducibility studies, SOPs, formulation/process variable mapping
Materials: Bio-based polymers, polymer thin films, organic materials, coatings, surface functionalization, electrochromic materials, adhesive/rheology ramp
Data & Workflow: Python/basic, JMP/basic, Excel, structured experimental records, data interpretation, high-throughput workflow concepts
Communication & Operations: Technical reports, presentations, publication-quality writing, lab operations, calibration systems, instrument uptime improvement

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." Chemistry of Materials, 2025, 37, 5558-5568.
2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." RSC Advances, 2018, 8, 20048-20055.
3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.
Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.
Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.
Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)